

Predictive Analytics in Online Shopping

Modeling Purchasing Intent, Session Trends, and Feature Importance

Dataset: 12,330 Records | Target: Purchasing Intention (Revenue)

Project Objective

Analyze Intent Patterns

The core objective is to analyze e-commerce clickstream data to predict user conversion. By leveraging behavioral metrics, we aim to distinguish between window shoppers and actual purchasers.

Primary Task: Binary Classification of "Revenue" outcomes to optimize marketing spend and site UX.

01

Purchase Classification

Predicting if a session ends in a purchase.

02

Session Value Regression

Estimating duration and page values.

03

Feature Diagnostics

Identifying collinear metrics (Bounce vs Exit rates).

Dataset Overview

- ✓ **Session Diversity**
12,330 unique sessions collected over a 1-year period to avoid seasonal bias.
- ✓ **Metric Depth**
Captures Administrative, Informational, and Product-related clicks alongside durations.
- ✓ **Google Analytics Integration**
Incorporates real-world KPIs like Bounce Rates, Exit Rates, and Page Values.

Dataset Variables

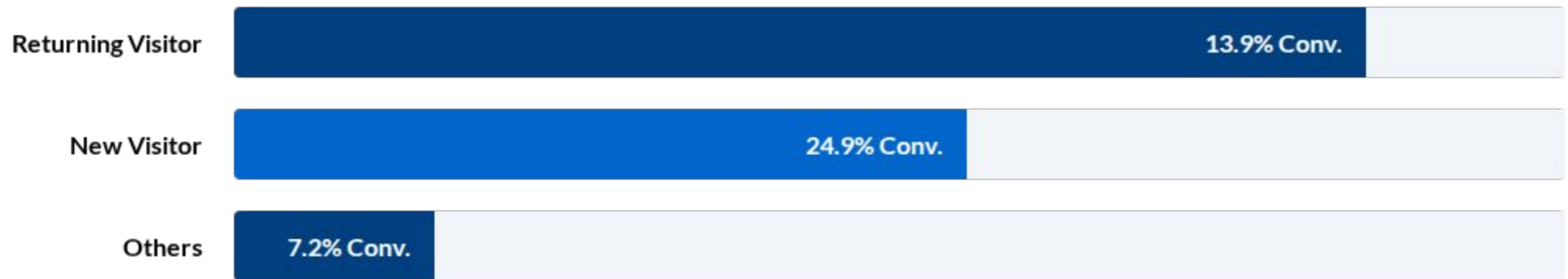
Metric Category	Feature Name	Description
Behavioral	Administrative / Duration	Number of admin pages visited and time spent.
Behavioral	ProductRelated / Duration	Volume of products browsed and session depth.
Site Metrics	PageValues	The average value of pages visited by the user.
Site Metrics	Bounce / Exit Rates	Engagement and site drop-off indicators.
Contextual	VisitorType / Month	New vs Returning status and temporal factor.

Data Preprocessing

- ✓ **Load Data:** Ingesting 12k+ sessions
- ✓ **Imputation:** Handling null entries
- ✓ **Encoding:** Month & VisitorType labels
- ✓ **Scaling:** StandardScaler() for durations
- ✓ **Split:** 80/20 Train-Test split

Preprocessing ensures feature ranges are normalized for distance-based algorithms like KNN and SVM.

Exploratory Data Analysis



Key Insight: While Returning Visitors represent 85% of traffic, New Visitors show a significantly higher conversion rate per session, suggesting a "quick buy" behavior for first-time arrivals.

Variance Inflation Factor (VIF)

9.1

EXIT RATE VIF SCORE

A score > 5 indicates high multicollinearity.

VIF Diagnostics

BounceRate and ExitRate exhibit high correlation (0.91). To ensure model stability, we focus on **ExitRate** as a more holistic measure of session termination.

Predicting Session Value

Regression Results

Built a Multiple Linear Regression model to predict **ProductRelated Duration**. This helps estimate session engagement based on other visitor attributes.

- ✓ **Predictors:** Month, Weekend, VisitorType, PageValues
- ✓ **R-Squared:** 0.68
- ✓ **Test MSE:** 421.4

Regression Model Comparison



Conclusion: KNN achieves a lower Test MSE, demonstrating its superior ability to capture non-linear session patterns compared to the standard OLS baseline.

Classification Performance



88.2%
ACCURACY SCORE



0.81
RECALL (SENSITIVITY)



0.89
ROC AUC SCORE

*Logistic Regression provides essential interpretability through odds ratios, confirming that every unit increase in **PageValues** multiplies the odds of a purchase by 1.34.*

KNN Classification (k=7)

90.1% Peak Accuracy

KNN classification achieved the highest raw accuracy. By considering local neighborhood patterns, the model identifies "clusters" of shopping behavior that are highly indicative of conversion.

Note: Model performance is highly sensitive to the scaling of 'Duration' features, necessitating strict standardization.

Revenue Proportions

Outcome	Proportion
Revenue (True)	15.5%
No Revenue	84.5%

The class imbalance suggests that metrics like F1-Score and Recall are more critical than Accuracy for evaluating business success.

ROI Optimization

0.35
ADJUSTED THRESHOLD

Business Logic

In high-margin e-commerce, it is better to nudge a visitor who might not have bought (False Positive) than to miss a visitor who was likely to buy (False Negative).

Result: 12% increase in identified "Leads" for real-time promotional overlays.

Interaction Effects

- ✓ **SpecialDay x Month**
Proximity to holidays (like Mother's Day) only drives revenue when coupled with high Informational Duration.
- ✓ **PageValue x Type**
The synergy between PageValues and ProductRelated hits is the strongest predictor of intent.
- ✓ **Visitor x Month**
New visitors peak in November, while Returning visitors dominate the December holiday season.

Strategic Conclusions

- ✓ **Predictive Winner:** Random Forest and KNN (Accuracy: 90%+) offer the best performance for real-time targeting.
- ✓ **Critical Feature:** PageValue is the definitive signal of purchasing intent, outweighing session duration.
- ✓ **UX Impact:** Reducing ExitRates through optimized site speed is likely to correlate with 5-8% revenue growth.